

1. The goal of this exercise is to make sure that you understand the notion of an interpretation.
 1. How many elements does the smallest domain of an interpretation contain?
 2. Can an interpretation domain be infinite?
 3. In an interpretation $\zeta = (\delta, \varphi)$ and for a concept name A , how many elements can/must $\varphi(A)$ have?
 4. In an interpretation $\zeta = (\delta, \varphi)$ and for a role name r , how many pairs of elements can/must $\varphi(r)$ have?
 5. For an element $e \in \delta$, can it be the case that $(e, e) \in \varphi(r)$?
 6. For two elements $e, f \in \delta$, can it be the case that $\{(e, f), (f, e)\} \subseteq \varphi(r)$?

Solution:

1. One element
2. Yes
3. Any number between 0 and the number of elements in δ
4. Any number between 0 and the number of elements in $\delta \times \delta$
5. Yes
6. Yes

2. Translate the following sentences into a DL sentence, using only the concept names:

Person, Happy, Unhappy, Animal, Cat, Old, Fish

and the role name *:Owns*. To make the current DL more expressive, let n be the positive integer, r be a role/relation, C be a concept, and $[\text{EXISTS } n \text{ } r \text{ } C]$ be a concept with a meaning “there are at least n role fillers for the role r that are instances of the concept C ”

1. happy person
2. happy pet owner
3. person who owns only cats
4. unhappy pet owners who own an old cat
5. pet owners who only own cats or fish

Solution:

1. $[\text{AND Person Happy}]$
2. $[\text{AND Person } [\text{EXISTS } 1 \text{ :Owns Animal}] \text{ Happy}]$

3. [AND Person [ALL :Owns Cat]]
4. [AND Person [EXISTS 1 :Owns Animal] Unhappy [EXISTS 1 :Owns [AND Cat Old]]]
5. [AND Person [ALL :Owns [OR Cat Fish]]]

3. Translate the following sentences into DL sentences, using only the concept names:

Student, HogwartsStudent, Hogwarts, Owl, Cat, Toad, Phoenix, Wizard

and the role name:

:AttendSchool, :HasPet.

Please use the definition of [EXISTS ...] as defined in the previous exercise, and [OR $c_1, \dots, c_n$] as $\varphi(c_1) \cup \dots \cup \varphi(c_n)$ with $c_1, \dots, c_n \in C$. Please also note that some sentences require **multiple DL sentences**.

1. Hogwarts students are exactly those students who attend Hogwarts school.
2. All pets of a Hogwarts student must be either an owl, a cat, or a toad.
3. Those who have phoenix as their pet is a wizard.
4. A phoenix is neither an owl, a cat, nor a toad.

Solution:

1. $\text{HogwartsStudent} \doteq [\text{AND Student } [\text{EXISTS 1 :AttendSchool Hogwarts}]]$
 2. $\text{HogwartsStudent} \sqsubseteq [\text{ALL :HasPet } [\text{OR Owl Cat Toad}]]$
 3. $[\text{EXISTS 1 :HasPet Phoenix}] \sqsubseteq \text{Wizard}$
 4. $[\text{AND Phoenix Owl}] \sqsubseteq \perp$
 5. $[\text{AND Phoenix Cat}] \sqsubseteq \perp$
 6. $[\text{AND Phoenix Toad}] \sqsubseteq \perp$
- (4) - (6) are representing the fourth sentence.

4. List all sentences that can be inferred if the following facts are added:

1. {HogwartsStudent(harryPotter)}
2. {HogwartsStudent(harryPotter), :HasPet(harryPotter, hedwig)}
3. {HogwartsStudent(harryPotter), :HasPet(harryPotter, hedwig), Phoenix(fawks)}
4. {HogwartsStudent(harryPotter), :HasPet(harryPotter, hedwig), Phoenix(fawks), :HasPet(dumbledore, fawks)}

Solution:

Bigger numbers cover all inferences from the smaller numbers, except the fourth answer that only covers all inferences from (1) and (2).

1. $[\text{EXISTS } 1 \text{ :AttendSchool Hogwarts}](\text{harryPotter})$
2. $[\text{OR Owl Cat Toad}](\text{hedwig})$
3. $\{\text{Owl(fawks)} \rightarrow \perp, \text{Cat(fawks)} \rightarrow \perp, \text{Toad(fawks)} \rightarrow \perp, [\text{FILLS :HasPet fawks}] \rightarrow \perp\}$
4. $\{\text{Owl(fawks)} \rightarrow \perp, \text{Cat(fawks)} \rightarrow \perp, \text{Toad(fawks)} \rightarrow \perp, \text{Wizard(dumbledore), HogwartsStudent(dumbledore)} \rightarrow \perp\}$

$\text{Owl(fawks)} \rightarrow \perp$ (and other similar forms) represents $\neg \text{Owl(fawks)}$. Since $\neg$ is not introduced in the original DL, the concept of negation can not be used.

$[\text{FILLS :HasPet fawks}] \rightarrow \perp$ is only true when there is only one `:HasPet` instance (i.e. `:HasPet(harryPotter, hedwig)`). The moment `:HasPet(dumbledore, fawks)` is introduced (in the fourth sentence), $[\text{FILLS :HasPet fawks}] \rightarrow \perp$ is no longer true anymore (i.e. not satisfiable).